

CSS

SENG-358: Web Programming

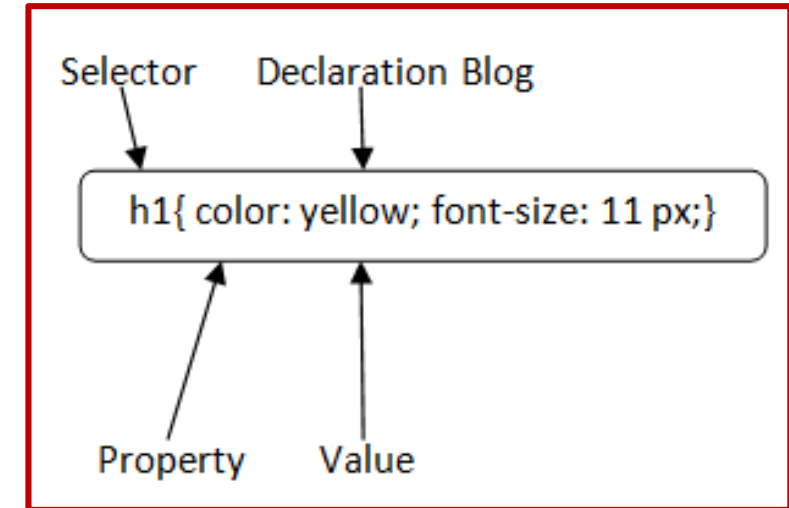
CSS

CSS

- the language we use to style an HTML document.
- describes how HTML elements should be displayed.

CSS Syntax

- A CSS rule set contains a selector and a declaration block.
- **Selector:** Selector indicates the HTML element you want to style. It could be any tag like `<h1>`, `<title>` etc.
- **Declaration Block:** The declaration block can contain one or more declarations separated by a semicolon.



CSS Selector

- CSS selectors are used to select the content you want to style. Selectors are the part of CSS rule set. CSS selectors select HTML elements according to its id, class, type, attribute etc.
- There are several different types of selectors in CSS.
 - CSS Element Selector
 - CSS Id Selector
 - CSS Class Selector
 - CSS Universal Selector
 - CSS Group Selector

The CSS element Selector

- The element selector selects HTML elements based on the element name.

```
<!DOCTYPE html>
<html>
<head>
<style>
  p {
    text-align: center;
    color: red;
  }
</style>
</head>
<body>
  <p>Every paragraph will be affected by the style.</p>
  <p id="para1">Me too!</p>
  <p>And me!</p>
</body>
</html>
```

Every paragraph will be affected by the style.

Me too!

And me!

The CSS id Selector

- The id selector uses the id attribute of an HTML element to select a specific element.
- The id of an element is unique within a page, so the id selector is used to select one unique element!
- To select an element with a specific id, write a hash (#) character, followed by the id of the element.

```
<!DOCTYPE html>
<html>
<head>
<style>
  #para1
  {
    text-align: center;
    color: red;
  }
</style>
</head>
<body>

<p id="para1">Hello World!</p>
<p>This paragraph is not affected by the style.</p>

</body>
</html>
```

Hello World!

This paragraph is not affected by the style.

The CSS class Selector

- The class selector selects HTML elements with a specific class attribute.
- To select elements with a specific class, write a period (.) character, followed by the class name.

```
<!DOCTYPE html>
<html>
<head>
<style>
  .center
  {
    text-align: center;
    color: red;
  }
</style>
</head>
<body>

<h1 class="center">Red and center-aligned heading</h1>
<p class="center">Red and center-aligned paragraph.</p>

</body>
</html>
```

Red and center-aligned heading

Red and center-aligned paragraph.

The CSS class Selector

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<style>
```

```
  p.center
```

```
  {
```

```
    text-align: center;
```

```
    color: red;
```

```
  }
```

```
</style>
```

```
</head>
```

```
<body>
```

```
<h1 class="center">This heading will not be affected</h1>
```

```
<p class="center">This paragraph will be red and center-aligned.</p>
```

```
</body>
```

```
</html>
```

This heading will not be affected

This paragraph will be red and center-aligned.

The CSS class Selector

```
<!DOCTYPE html>
<html>
<head>
<style>
p.center {
  text-align: center;
  color: red;
}

p.large {
  font-size: 300%;
}
</style>
</head>
<body>
```

```
<h1 class="center">This heading will not be affected</h1>
<p class="center">This paragraph will be red and center-aligned.</p>
<p class="center large">This paragraph will be red, center-aligned, and in a large font-size.</p>

</body>
</html>
```

This heading will not be affected

This paragraph will be red and center-aligned.

This paragraph will be red, center-aligned, and in a large font-size.

The CSS Universal Selector

- The universal selector (*) selects all HTML elements on the page.

```
<!DOCTYPE html>
<html>
<head>
<style>
  *
  {
    text-align: center;
    color: blue;
  }
</style>
</head>
<body>

<h1>Hello world!</h1>

<p>Every element on the page will be affected by the style.</p>
<p id="para1">Me too!</p>
<p>And me!</p>

</body>
</html>
```

Hello world!

Every element on the page will be affected by the style.

Me too!

And me!

The CSS Grouping Selector

- The grouping selector selects all the HTML elements with the same style definitions.
- Look at the following CSS code (the h1, h2, and p elements have the same style definitions)

```
<!DOCTYPE html>
<html>
<head>
<style>
  h1, h2, p
  {
    text-align: center;
    color: red;
  }
</style>
</head>
<body>

<h1>Hello World!</h1>
<h2>Smaller heading!</h2>
<p>This is a paragraph.</p>

</body>
</html>
```

Hello World!

Smaller heading!

This is a paragraph.

Three Ways to Insert CSS

- External CSS
- Internal CSS
- Inline CSS

Inline CSS

- An inline style may be used to apply a unique style for a single element.
- To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

External CSS

- With an external style sheet, you can change the look of an entire website by changing just one file!
- Each HTML page must include a reference to the external style sheet file inside the <link> element, inside the head section.



```

1  <!DOCTYPE html>
2  <html lang="en">
3  <head>
4      <meta charset="UTF-8">
5      <meta http-equiv="X-UA-Compatible" content="IE=edge">
6      <meta name="viewport" content="width=device-width, initial-scale=1.0">
7      <title>Document</title>
8      <link rel="stylesheet" href="mystyle.css">
9  </head>
10 <body>
11     <h1>This is a heading</h1>
12     <p>This is a paragraph.</p>
13 </body>
14 </html>
```

Internal CSS

- An internal style sheet may be used if one single HTML page has a unique style.
- The internal style is defined inside the <style> element, inside the head section.

```
<!DOCTYPE html>
<html>
<head>
<style>
  body
  {
    background-color: linen;
  }
  h1
  {
    color: maroon;
    margin-left: 40px;
  }
</style>
</head>
<body>
<h1>This is a heading</h1>
<p>This is a paragraph.</p>

</body>
</html>
```

This is a heading

This is a paragraph.

Inline CSS

- An inline style may be used to apply a unique style for a single element.
- To use inline styles, add the style attribute to the relevant element. The style attribute can contain any CSS property.

Inline CSS

- Inline styles are defined within the *"style" attribute* of the relevant element.

```
<!DOCTYPE html>
```

```
<html>
```

```
<body>
```

```
<h1 style="color:blue;text-align:center;">This is a heading</h1>
```

```
<p style="color:red;">This is a paragraph.</p>
```

```
</body>
```

```
</html>
```

This is a heading

This is a paragraph.

Multiple Style Sheets

- If some properties have been defined for the same selector (element) in different style sheets, the value from the last read style sheet will be used.

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" type="text/css" href="mystyle.css">
<style>
  h1
  {
    color: orange;
  }
</style>
</head>
<body>
  <h1>This is a heading</h1>
  <p>The style of this document is a combination of an external stylesheet, and
internal style</p>
</body>
</html>
```

This is a heading

The style of this document is a combination of an external stylesheet, and internal style

CSS Attribute Selectors

- **CSS [attribute] Selector**
- The [attribute] selector is used to select elements with a specified attribute.
- The following example selects all <a> elements with a target attribute:

```
<!DOCTYPE html>
<html>
<head>
<style>
a[target] {
  background-color: yellow;
}
</style>
</head>
<body>
```

CSS [attribute] Selector

The links with a target attribute gets a yellow background:

[w3schools.com](https://www.w3schools.com) [disney.com](http://www.disney.com) [wikipedia.org](http://www.wikipedia.org)

```
<h2>CSS [attribute] Selector</h2>
<p>The links with a target attribute gets a yellow background:</p>

<a href="https://www.w3schools.com">w3schools.com</a>
<a href="http://www.disney.com" target="_blank">disney.com</a>
<a href="http://www.wikipedia.org" target="_top">wikipedia.org</a>

</body>
</html>
```

CSS Attribute Selectors

- **CSS [attribute="value"] Selector**
- The [attribute="value"] selector is used to select elements with a specified attribute and value.
- The following example selects all <a> elements with a target="_blank" attribute: CSS [attribute="value"] Selector

```
<!DOCTYPE html>
<html>
<head>
<style>
a[target="_blank"] {
  background-color: yellow;
}
</style>
</head>
<body>

<h2>CSS [attribute="value"] Selector</h2>
<p>The link with target="_blank" gets a yellow background:</p>

<a href="https://www.w3schools.com">w3schools.com</a>
<a href="http://www.disney.com" target="_blank">disney.com</a>
<a href="http://www.wikipedia.org" target="_top">wikipedia.org</a>
```

CSS [attribute="value"] Selector

The link with target="_blank" gets a yellow background:

[w3schools.com](https://www.w3schools.com) [disney.com](http://www.disney.com) [wikipedia.org](http://www.wikipedia.org)

CSS Attribute Selectors

- **CSS [attribute~="value"] Selector**
- The [attribute~="value"] selector is used to select elements with an attribute value containing a specified word.
- The following example selects all elements with a title attribute that contains a word **"flower"**

```
<!DOCTYPE html>
<html>
<head>
<style>
[title~="flower"] {
    border: 5px solid yellow;
}
</style>
</head>
<body>
<h2>CSS [attribute~="value"] Selector</h2>
<p>All images with the title attribute containing the word "flower" get a yellow border.
</p>



</body>
</html>
```

CSS Attribute Selectors

- **CSS [attribute~="value"] Selector**
- The [attribute~="value"] selector is used to select elements with an attribute value containing a specified word.
- The following example selects all elements with a title attribute that contains a word "flower".

CSS [attribute~="value"] Selector

All images with the title attribute containing the word "flower" get a yellow border.



CSS Attribute Selectors

▪ CSS [attribute]="value" Selector

- The [attribute]="value" selector is used to select elements with the specified attribute, whose value can be exactly the specified value, or the specified value followed by a hyphen (-).
- Note: The value has to be a whole word, either alone, like class="top", or followed by a hyphen(-), like class="top-text".

```
<!DOCTYPE html>
<html>
<head>
<style>
[class="top"] {
  background: yellow;
}
</style>
</head>
<body>

<h2>CSS [attribute]="value" Selector</h2>

<h1 class="top-header">Welcome</h1>
<p class="top-text">Hello world!</p>
<p class="topcontent">Are you learning CSS?</p>
```

CSS [attribute]="value" Selector

Welcome

Hello world!

Are you learning CSS?

CSS Attribute Selectors

- **CSS [attribute^="value"] Selector**
- The [attribute^="value"] selector is used to select elements with the specified attribute, whose value starts with the specified value.
- The following example selects all elements with a class attribute **value that starts with "top"**

```
<style>
[class^="top"] {
  background: yellow;
}
</style>
</head>
<body>

<h2>CSS [attribute^="value"] Selector</h2>

<h1 class="top-header">Welcome</h1>
<p class="top-text">Hello world!</p>
<p class="topcontent">Are you learning CSS?</p>

</body>
</html>
```

CSS [attribute^="value"] Selector

Welcome

Hello world!

Are you learning CSS?

CSS Attribute Selectors

- **CSS [attribute\$="value"] Selector**
- The [attribute\$="value"] selector is used to select elements whose attribute value ends with a specified value.
- The following example selects all elements with a class attribute **value that ends with "test"**

```
<style>
[class$="test"] {
  background: yellow;
}
</style>
</head>
<body>

<h2>CSS [attribute$="value"] Selector</h2>

<div class="first_test">The first div element.</div>
<div class="second">The second div element.</div>
<div class="my-test">The third div element.</div>
<p class="mytest">This is some text in a paragraph.</p>
```

CSS [attribute\$="value"] Selector

The first div element.

The second div element.

The third div element.

This is some text in a paragraph.

CSS Attribute Selectors

- **CSS [attribute*="value"] Selector**
- The [attribute*="value"] selector is used to select elements whose attribute value contains a specified value.
- The following example selects all elements with a class attribute **value that contains "te"**

```
<style>
[class*="te"] {
  background: yellow;
}
</style>
</head>
<body>

<h2>CSS [attribute*="value"] Selector</h2>

<div class="first_test">The first div element.</div>
<div class="second">The second div element.</div>
<div class="my-test">The third div element.</div>
<p class="mytest">This is some text in a paragraph.</p>
```

CSS [attribute*="value"] Selector

The first div element.

The second div element.

The third div element.

This is some text in a paragraph.

CSS Pseudo-classes

- **Pseudo-classes:** A pseudo-class is used to define a special state of an element.
- it can be used to:
 - Style an element when a user mouses over it
 - Style visited and unvisited links differently
 - Style an element when it gets focus
- Syntax
- The syntax of pseudo-classes:

```
selector:pseudo-class {  
    property: value;  
}
```

CSS Pseudo-classes

▪ Anchor Pseudo-classes

```
<style>
/* unvisited link */
a:link {
  color: red;
}

/* visited link */
a:visited {
  color: green;
}

/* mouse over link */
a:hover {
  color: hotpink;
}

/* selected link */
a:active {
  color: blue;
}
</style>
</head>
<body>
<h2>Styling a link depending on state</h2>
<p><b><a href="default.asp" target="_blank">This is a link</a></b></p>
</body>
</html>
```

Styling a link depending on state

[This is a link](#)

CSS Pseudo-classes

- Pseudo-classes and HTML Classes

```
<style>
a.highlight:hover {
  color: #ff0000;
  font-size: 22px;
}
</style>
</head>
<body>

<h2>Pseudo-classes and HTML Classes</h2>

<p>When you hover over the first link below, it will change color and font size:</p>

<p><a class="highlight" href="css_syntax.asp">CSS Syntax</a></p>

<p><a href="default.asp">CSS Tutorial</a></p>
```

Pseudo-classes and HTML Classes

When you hover over the first link below, it will change color and font size:

[CSS Syntax](#)

[CSS Tutorial](#)

CSS Pseudo-classes

- **Hover on <div>**

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: green;
  color: white;
  padding: 25px;
  text-align: center;
}

div:hover {
  background-color: blue;
}
</style>
</head>
<body>

<p>Mouse over the div element below to change its background color:</p>

<div>Mouse Over Me</div>
```

Mouse over the div element below to change its background color:

Mouse Over Me

CSS Pseudo-classes

- **Simple Tooltip Hover**
- Hover over a `<div>` element to show a `<p>` element (like a tooltip)

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  display: none;
  background-color: yellow;
  padding: 20px;
}
```

```
div:hover p {
  display: block;
}
</style>
</head>
<body>
```

```
<div>Hover over this div element to show the p element
  <p>Tada! Here I am!</p>
</div>
```

Hover over this div element to show the p element

Tada! Here I am!

CSS Pseudo-classes

- **CSS - The : first-child Pseudo-class**
- The : first-child pseudo-class matches a specified element that is the first child of another element.

```
<head>
<style>
p:first-child {
  color: blue;
}
</style>
</head>
<body>

<p>This is some text.</p>           This is some text.
<p>This is some text.</p>         This is some text.

<div>
  <p>This is some text.</p>         This is some text.
  <p>This is some text.</p>
</div>                             This is some text.

</body>
</html>
```

CSS Pseudo-classes

- **Match the first `<i>` element in all `<p>` elements**
- In the following example, the selector matches the first `<i>` element in all `<p>` elements.

```
<!DOCTYPE html>
<html>
<head>
<style>
p i:first-child {
  color: blue;
}
</style>
</head>
<body>

<p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>
<p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>

</body>
</html>
```

I am a *strong* person. I am a *strong* person.

I am a *strong* person. I am a *strong* person.

CSS Pseudo-classes

- **Match all `<i>` elements in all first child `<p>` elements**
- In the following example, the selector matches all `<i>` elements in `<p>` elements that are the first child of another element.

```
<head>
<style>
p:first-child i {
  color: blue;
}
</style>
</head>
<body>

<p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>
<p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>

<div>
  <p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>
  <p>I am a <i>strong</i> person. I am a <i>strong</i> person.</p>
</div>
```

I am a *strong* person. I am a *strong* person.

I am a *strong* person. I am a *strong* person.

I am a *strong* person. I am a *strong* person.

I am a *strong* person. I am a *strong* person.

YOU CAN USE THE FIRST-LINE PSEUDO-ELEMENT TO ADD A SPECIAL EFFECT TO THE
first line of a text. Some more text. And even more, and more, and more, and
more, and more, and more, and more, and more, and more, and more, and more,
and more.

CSS Pseudo-elements

- **The ::first-letter Pseudo-element**
- The ::first-letter pseudo-element is used to add a special style to the first letter of a text.
- The following example formats the first letter of the text in all <p> elements.

```
<head>
<style>
p.intro::first-letter {
  color: #ff0000;
  font-size: 200%;
}
</style>
</head>
<body>
```

```
<p class="intro">This is an introduction.</p>
<p>This is a paragraph with some text. A bit more text even.</p>

</body>
</html>
```

This is an introduction.

This is a paragraph with some text. A bit more text even.

CSS Pseudo-elements

▪ Multiple Pseudo-elements

- Several pseudo-elements can also be combined. In the following example, the first letter of a paragraph will be red, in an xx-large font size. The rest of the first line will be blue, and in small-caps. The rest of the paragraph will be the default font size and color:

```
<!DOCTYPE html>
<html>
<head>
<style>
p::first-letter {
  color: #ff0000;
  font-size: xx-large;
}

p::first-line {
  color: #0000ff;
  font-variant: small-caps;
}
</style>
</head>
<body>
<p>You can combine the ::first-letter and ::first-line pseudo-elements to add a special
effect to the first letter and the first line of a text!</p>
```

YOU CAN COMBINE THE ::FIRST-LETTER AND ::FIRST-LINE PSEUDO-ELEMENTS TO ADD A SPECIAL EFFECT TO THE first letter and the first line of a text!

CSS Pseudo-elements

- **CSS - The ::before Pseudo-element**
- The ::before pseudo-element can be used to insert some content before the content of an element. The following example inserts an image before the content of each <h1> element

```
<!DOCTYPE html>
<html>
<head>
<style>
h1::before {
  content: url(smiley.gif);
}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>The ::before pseudo-element inserts content before the content of an element.</p>

<h1>This is a heading</h1>

</body>
</html>
```



This is a heading

The ::before pseudo-element inserts content before the content of an element.



This is a heading

CSS Pseudo-elements

- **CSS - The ::after Pseudo-element**
- The ::after pseudo-element can be used to insert some content after the content of an element. The following example inserts an image after the content of each <h1> element:

```
<!DOCTYPE html>
<html>
<head>
<style>
h1::after {
  content: url(smiley.gif);
}
</style>
</head>
<body>

<h1>This is a heading</h1>
<p>The ::after pseudo-element inserts content after the content of an element.</p>

<h1>This is a heading</h1>
```

This is a heading

The ::after pseudo-element inserts content after the content of an element.

This is a heading

CSS Pseudo-elements

- **CSS - The ::marker Pseudo-element**
- The ::marker pseudo-element **selects the markers of list items**. The following example styles the markers of list items:

```
<head>
<style>
  ::marker {
    color: red;
    font-size: 23px;
  }
</style>
</head>
<body>
```

```
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Milk</li>
</ul>
```

- Coffee
- Tea
- Milk

```
<ol>
  <li>First</li>
  <li>Second</li>
  <li>Third</li>
</ol>
```

1. First
2. Second
3. Third

CSS Pseudo-elements

▪ CSS - The ::selection Pseudo-element

- The ::selection pseudo-element matches the portion of an element that is selected by a user.
- The following CSS properties can be applied to ::selection: color, background, cursor, and outline.

```
<head>
<style>
::selection {
  color: red;
  background: yellow;
}
</style>
</head>
<body>
```

```
<h1>Select some text on this page:</h1>

<p>This is a paragraph.</p>
<div>This is some text in a div element.</div>
```

Select some text on this page:

This is a paragraph.

This is some text in a div element.

CSS Colors

- **CSS Background Color**

- You can set the background color for HTML elements.

- **CSS Text Color**

- You can set the color of text.

- **CSS Border Color**

- You can set the color of borders.

- **CSS Color Names:** In CSS, a color can be specified by using a [predefined color name](#).

- **CSS Color Values:** In CSS, [colors can also be specified using RGB values, HEX values, HSL values, RGBA values, and HSLA values](#): Same as color name "Tomato".

CSS Colors

▪ Example

```
<!DOCTYPE html>
<html>
<body>

<h1 style="border: 2px solid Tomato;">Border Color</h1>
<h1 style="border: 2px solid DodgerBlue;">Border Color</h1>
<h1 style="border: 2px solid Violet;">Border Color</h1>

<h3 style="color:Tomato;">Header Text Color</h3>

<p style="color:DodgerBlue;">Paragraph Text Color</p>

<h1 style="background-color:DodgerBlue;">Header Background Color</h1>

</body>
</html>
```

Border Color

Border Color

Border Color

Header Text Color

Paragraph Text Color

Header Background Color

CSS Colors

▪ Example

```
<!DOCTYPE html>
<html>
<body>

<h1 style="background-color:Tomato;">Tomato</h1>
<h1 style="background-color:Orange;">Orange</h1>
<h1 style="background-color:rgb(255, 99, 71);">rgb(255, 99, 71)</h1>
<h1 style="background-color:#ff6347;">#ff6347</h1>
<p>Same as color name "Tomato", but 50% transparent:</p>
<h1 style="background-color:rgba(255, 99, 71, 0.5);">rgba(255, 99, 71, 0.5)</h1>
<h1 style="background-color:hsla(9, 100%, 64%, 0.5);">hsla(9, 100%, 64%, 0.5)</h1>

</body>
</html>
```

Tomato

Orange

rgb(255, 99, 71)

#ff6347

Same as color name "Tomato", but 50% transparent:

rgba(255, 99, 71, 0.5)

hsla(9, 100%, 64%, 0.5)

CSS Colors

- **CSS RGB Colors:** An RGB color value represents RED, GREEN, and BLUE light sources.
- **RGB Value**
- In CSS, a color can be specified as an RGB value, using this formula: **rgb(red, green, blue)**
- **RGBA Value**
- RGBA color values are an extension of RGB color values with an alpha channel - which specifies the **opacity for a color**.
- **CSS HEX Colors:** A hexadecimal color is specified with: **#RRGGBB**, where the **RR (red)**, **GG (green)** and **BB (blue)** hexadecimal integers specify the components of the color.
- **HEX Value**
- In CSS, a color can be specified using a hexadecimal value in the form: **#rrggbb**

CSS Colors

- **CSS HSL Colors:** HSL stands for hue, saturation, and lightness.
- **HSL Value**
- In CSS, a color can be specified using hue, saturation, and lightness (HSL) in the form: **hsl(hue, saturation, lightness)**

```
<!DOCTYPE html>
<html>
<body>

<h1>Specify colors using RGB values</h1>

<h2 style="background-color:rgb(255, 0, 0);">rgb(255, 0, 0)</h2>

<h1>Specify colors using HEX values</h1>

<h2 style="background-color:#ffa500;">#ff0000</h2>

<h1>Specify colors using HSL values</h1>

<h2 style="background-color:hsl(300, 76%, 72%);">hsl(0, 100%, 50%)</h2>

</body>
</html>
```

Specify colors using RGB values

rgb(255, 0, 0)

Specify colors using HEX values

#ff0000

Specify colors using HSL values

hsl(0, 100%, 50%)

CSS Backgrounds

- The CSS background properties are used to add background effects for elements.
- CSS background properties:
 - background-color
 - background-image
 - background-repeat
 - background-attachment
 - background-position
 - background (shorthand property)

CSS Backgrounds

- **background-color:** The background-color property specifies the background color of an element.
- **Opacity / Transparency:** The opacity property specifies the opacity/transparency of an element. It can take a value from 0.0 - 1.0. The lower value, the more transparent.
- **CSS background-image:** The background-image property specifies an image to use as the background of an element.

CSS Borders

- The CSS border properties are use to specify the style, color and size of the border of an element.
- The CSS border properties are given below
 - border-style
 - border-color
 - border-width
 - border-radius

CSS Borders

- **CSS border-style:**

The Border style property is used to specify the border type which you want to display on the web page.

Value	Description
none	It doesn't define any border.
dotted	It is used to define a dotted border.
dashed	It is used to define a dashed border.
solid	It is used to define a solid border.
double	It defines two borders with the same border-width value.
groove	It defines a 3d grooved border. effect is generated according to border-color value.
ridge	It defines a 3d ridged border. effect is generated according to border-color value.
inset	It defines a 3d inset border. effect is generated according to border-color value.
outset	It defines a 3d outset border. effect is generated according to border-color value.

CSS Borders

- **CSS border-width:** The border-width property is used to set the border's width. It is set in pixels.
- **CSS border-color:** There are three methods to set the color of the border.
- The color can be set by:
 - **name** - specify a color name, like "red"
 - **HEX** - specify a HEX value, like "#ff0000"
 - **RGB** - specify a RGB value, like "rgb(255,0,0)"
 - **HSL** - specify a HSL value, like "hsl(0, 100%, 50%)"
 - **transparent**

CSS Borders

▪ Example:

```
<style>
p.one {
  border-style: solid;
  border-width: 8px;
  border-color: black;
}

p.two {
  border-style: dashed;
  border-color: green;
}

p.three {
  border-style: dotted;
  border-color: blue;
}
</style>
</head>
<body>

<h2>Border-style, border-width and border-color</h2>

<p class="one">A solid red border</p>
<p class="two">A dashed green border</p>
<p class="three">A dotted blue border</p>
```

Border-style, border-width and border-color

A solid red border

A solid green border

A dotted blue border

CSS Borders

- **CSS Border - Shorthand Property:** Like you saw in the previous page, there are many properties to consider when dealing with borders.
- To shorten the code, it is also possible to specify all the individual border properties in one property.
- The border property is a shorthand property for the following individual border properties:
 - border-width
 - border-style (required)
 - border-color

CSS Borders

- **CSS Rounded Borders:** The border-radius property is used to add rounded borders to an element:

Normal border

Round border

Rounder border

Roudest border

CSS Borders

▪ Example:

```
<style>
p.normal {border: 2px solid red; padding: 5px;}

p.round1 {border: 3px dashed black; border-radius: 5px; padding: 5px;}

p.round2 {border: 2px solid green; border-radius: 8px; padding: 5px;}

p.round3 {border: 2px solid blue; border-radius: 12px; padding: 5px;}
</style>
</head>
<body>

<h2>The border-radius Property</h2>
<p>This property is used to add rounded borders to an element:</p>

<p class="normal">Normal border</p>
<p class="round1">Round border</p>
<p class="round2">Rounder border</p>
<p class="round3">Roundest border</p>

</body>
</html>
```

The border-radius Property

This property is used to add rounded borders to an element:

Normal border

Round border

Rounder border

Roundest border

CSS Margins

- CSS Margin property is used to define the space around elements. It is completely transparent and doesn't have any background color. It clears an area around the element.
- CSS Margin Properties

Property	Description
<u>margin</u>	A shorthand property for setting all the margin properties in one declaration
<u>margin-bottom</u>	Sets the bottom margin of an element
<u>margin-left</u>	Sets the left margin of an element
<u>margin-right</u>	Sets the right margin of an element
<u>margin-top</u>	Sets the top margin of an element

CSS Margins

- CSS has properties for specifying the margin for each side of an element:
 - `margin-top`
 - `margin-right`
 - `margin-bottom`
 - `margin-left`
- All the margin properties can have the following values:
 - **auto:** The browser calculates the margin: You can set the margin property to `auto` to horizontally center the element within its container.
 - **length:** It specifies a margin in px, pt, cm, etc.
 - **%:** It specifies a margin in % of the width of the containing element
 - **inherit:** It specifies that the margin should be inherited from the parent element

CSS Margins

- CSS has properties for specifying the margin for each side of an element:
 - `margin-top`
 - `margin-right`
 - `margin-bottom`
 - `margin-left`
- All the margin properties can have the following values:
 - **auto:** The browser calculates the margin: You can set the margin property to `auto` to horizontally center the element within its container.
 - **length:** It specifies a margin in px, pt, cm, etc.
 - **%:** It specifies a margin in % of the width of the containing element
 - **inherit:** It specifies that the margin should be inherited from the parent element

CSS Margins

▪ Example

```
<head>
<style>
div {
  border: 1px solid black;
  margin-top: 100px;
  margin-bottom: 100px;
  margin-right: 150px;
  margin-left: 80px;
  background-color: lightgray;
}
</style>
</head>
<body>

<h2>Using individual margin properties</h2>

<div>This div element has a top margin of 100px, a right margin of 150px, a bottom
margin of 100px, and a left margin of 80px.</div>

</body>
</html>
```

Using individual margin properties

This div element has a top margin of 100px, a right margin of 150px, a bottom margin of 100px, and a left margin of 80px.

CSS Margins

- **Margin - Shorthand Property:**

To shorten the code, it is possible to specify all the margin properties in one property.

- The margin property is a shorthand property for the following individual margin properties:

- margin-top
- margin-right
- margin-bottom
- margin-left

```
p {  
  margin: 25px 50px 75px 100px;  
}
```

top margin is 25px
right margin is 50px
bottom margin is 75px
left margin is 100px

```
p {  
  margin: 25px 50px 75px;  
}
```

top margin is 25px
right and left margin are 50px
bottom margin is 75px

```
p {  
  margin: 25px 50px;  
}
```

top and bottom margin are 25px
right and left margin are 50px

CSS Padding

- The CSS padding properties are used to generate space around an element's content, inside of any defined borders.
- **All CSS Padding Properties**

Property	Description
<u>padding</u>	A shorthand property for setting all the padding properties in one declaration
<u>padding-bottom</u>	Sets the bottom padding of an element
<u>padding-left</u>	Sets the left padding of an element
<u>padding-right</u>	Sets the right padding of an element
<u>padding-top</u>	Sets the top padding of an element

CSS Padding

▪ Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  border: 1px solid black;
  background-color: lightblue;
  padding-top: 50px;
  padding-right: 30px;
  padding-bottom: 50px;
  padding-left: 80px;
}
h2 {padding: 50px 30px 50px 80px; border: 1px solid black;}
</style>
</head>
<body>

<h2>Padding - Shorthand Property</h2>

<div>This div element has a top padding of 50px, a right padding of 30px, a bottom
padding of 50px, and a left padding of 80px.</div>

</body>
</html>
```

Padding - Shorthand Property

This div element has a top padding of 50px, a right padding of 30px, a bottom padding of 50px, and a left padding of 80px.

CSS Height, Width and Max-width

- The CSS height and width properties are used to set the height and width of an element.
- The CSS max-width property is used to set the maximum width of an element.

Property	Description
<u>height</u>	Sets the height of an element
<u>max-height</u>	Sets the maximum height of an element
<u>max-width</u>	Sets the maximum width of an element
<u>min-height</u>	Sets the minimum height of an element
<u>min-width</u>	Sets the minimum width of an element
<u>width</u>	Sets the width of an element

CSS Height, Width and Max-width

- **CSS height and width Values:** The height and width properties may have the following values:
 - **auto:** This is default. The browser calculates the height and width
 - **length:** Defines the height/width in px, cm, etc.
 - **%:** Defines the height/width in percent of the containing block
 - **initial:** Sets the height/width to its default value
 - **inherit:** The height/width will be inherited from its parent value

CSS Height, Width and Max-width

▪ Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  height: 200px;
  width: 50%;
  background-color: lightblue;
}
</style>
</head>
<body>

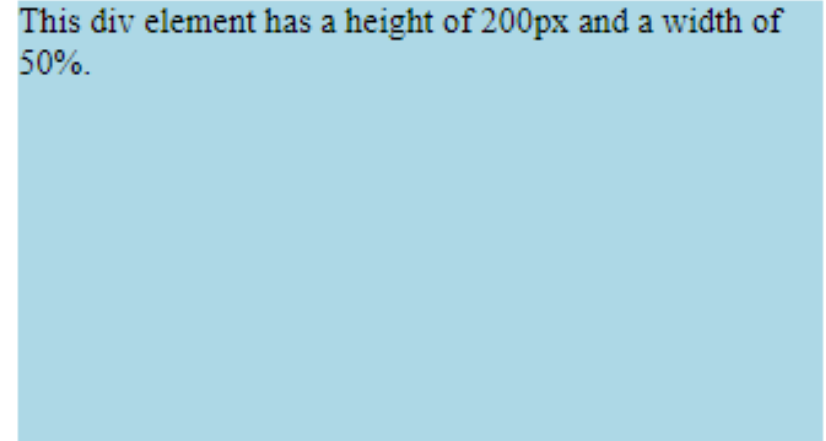
<h2>Set the height and width of an element</h2>

<div>This div element has a height of 200px and a width of 50%.</div>

</body>
</html>
```

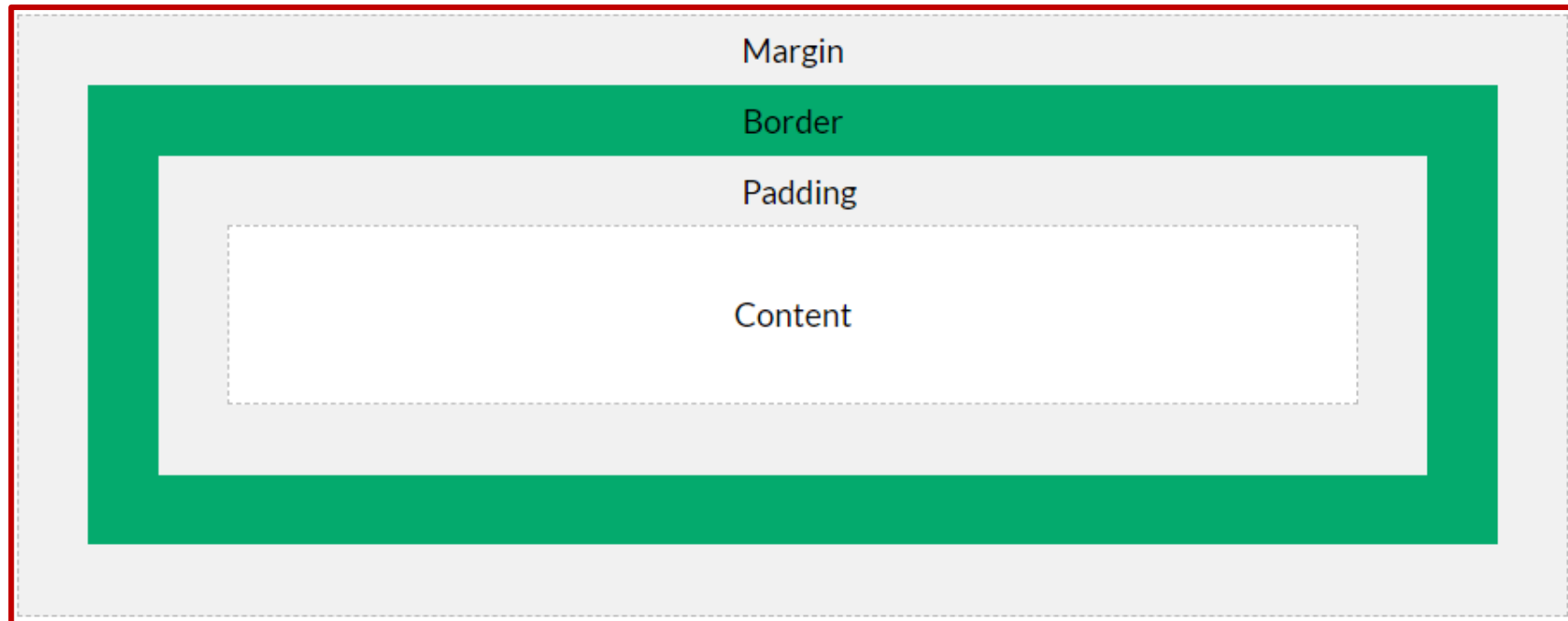
Set the height and width of an element

This div element has a height of 200px and a width of 50%.



CSS Box Model

- In CSS, the term "box model" is used when talking about [design and layout](#).
- The CSS box model is essentially a box that wraps around every HTML element. It consists of: [margins, borders, padding, and the actual content](#). The image below illustrates the box model:



CSS Box Model

▪ Example

```
<!DOCTYPE html>
<html>
<head>
<style>
div {
  background-color: lightgrey;
  width: 300px;
  border: 15px solid green;
  padding: 50px;
  margin: 20px;
}
</style>
</head>
<body>

<h2>A Box is added to content</h2>

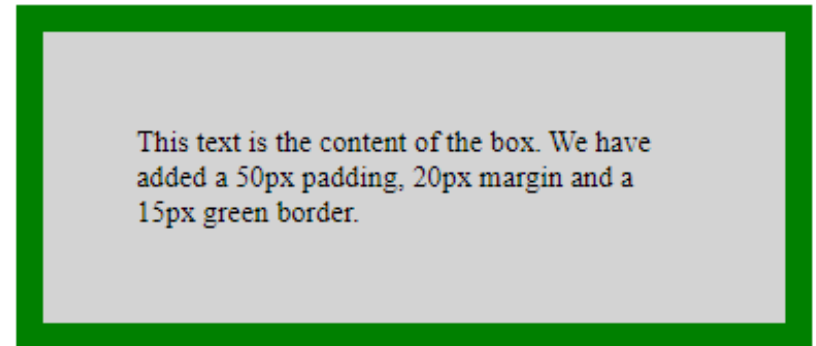
<p>The CSS box model is essentially a box that wraps around every HTML element.<br> It
consists of: borders, padding, margins, and the actual content.</p>

<div>This text is the content of the box. We have added a 50px padding, 20px margin and
a 15px green border. </div>

</body>
</html>
```

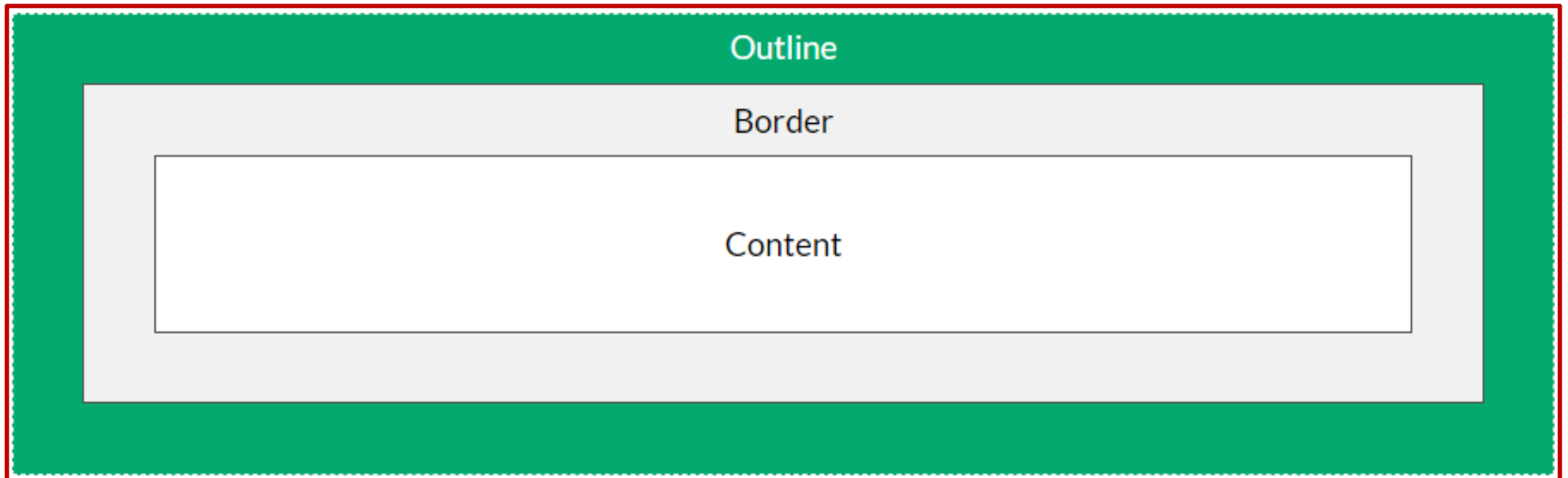
A Box is added to content

The CSS box model is essentially a box that wraps around every HTML element. It consists of: borders, padding, margins, and the actual content.



CSS Outline

- An outline is a line drawn outside the element's border.
- Unlike border, the outline is drawn outside the element's border, and may overlap other content. Also, the outline is NOT a part of the element's dimensions; the element's total width and height is not affected by the width of the outline.



CSS Outline

- CSS has the following outline properties:
 - **outline-style:** *dotted, dashed, solid, double, groove, ridge, inset, outset, none*
 - **hidden outline styles:** *thin (typically 1px), medium (typically 3px), thick (typically 5px), a specific size (in px, pt, cm, em, etc)*
 - **outline-color**
 - **outline-width**
 - **outline-offset:** *The outline-offset property adds space between an outline and the edge/border of an element. The space between an element and its outline is transparent.*

CSS Outline

▪ Example

```
<!DOCTYPE html>
<html>
<head>
<style>
p {
  margin: 30px;
  border: 1px solid black;
  outline: 1px solid red;
  outline-offset: 15px;
}
</style>
</head>
<body>

<h2>The outline-offset Property</h2>

<p>This paragraph has an outline 15px outside the border edge.</p>

</body>
</html>
```

The outline-offset Property

This paragraph has an outline 15px outside the border edge.

CSS Text

- CSS has a lot of properties for formatting text.
- **Text Color:** The color property is used to set the color of the text.
- **Text Alignment:** The text-align property is used to set the horizontal alignment of a text.
 - A text can be [left](#) or [right aligned](#), [centered](#), or [justified](#).
- **Text Direction:** The direction and unicode-bidi properties can be used to change the text direction of an element:
- **Vertical Alignment:** The vertical-align property sets the vertical alignment of an element.

CSS Text

▪ Example

```
<head>
<style>
h1 {
  text-align: center; color: #ff00ff;
}

h2 {
  text-align: left; color: rgb(200, 0, 199);
}

h3 {
  text-align: right; color: hsl(10, 50%, 66%);
}
</style>
</head>
<body>

<h1>Heading 1 (center)</h1>
<h2>Heading 2 (left)</h2>
<h3>Heading 3 (right)</h3>
```

Heading 1 (center)

Heading 2 (left)

Heading 3 (right)

The three headings above are aligned center, left and right.
with different colors.

```
<p>The three headings above are aligned center, left and right.<br>
with different colors.</p>
```

CSS Text

- The CSS Text Alignment/Direction Properties

Property	Description
<u>direction</u>	Specifies the text direction/writing direction
<u>text-align</u>	Specifies the horizontal alignment of text right, left, centered, justified
<u>text-align-last</u>	Specifies how to align the last line of a text
<u>unicode-bidi</u>	Used together with the <u>direction</u> property to set or return whether the text should be overridden to support multiple languages in the same document
<u>vertical-align</u>	Sets the vertical alignment of an element baseline, text-top, text-bottom, sub, super

CSS Text

▪ Example

```
<head>
<style>
.div_1 {
  border: 1px solid black;
  padding: 10px;
  width: 200px;
  height: 100px;
  text-align: justify;
}
.div_2 {
  border: 1px solid black;
  padding: 10px;
  width: 200px;
  height: 100px;
  text-align: left;
}
</style>
</head>
<body>

<h1>Justified Text</h1>
<div class="div_1">
In my younger and more vulnerable years my father gave me some advice that
turning over in my mind ever since.</div>
<h1>Left Aligned Text</h1>
<div class="div_2">
In my younger and more vulnerable years my father gave me some advice that
turning over in my mind ever since.</div>
```

Justified Text

In my younger and more vulnerable years my father gave me some advice that I've been turning over in my mind ever since.

Left Aligned Text

In my younger and more vulnerable years my father gave me some advice that I've been turning over in my mind ever since.

CSS Text

- All CSS text-decoration Properties

Property	Description
<u>text-decoration</u>	Sets all the text-decoration properties in one declaration
<u>text-decoration-color</u>	Specifies the color of the text-decoration
<u>text-decoration-line</u>	Specifies the kind of text decoration to be used (underline, overline, etc.)
<u>text-decoration-style</u>	Specifies the style of the text decoration (solid, dotted, etc.)
<u>text-decoration-thickness</u>	Specifies the thickness of the text decoration line

CSS Text

- **Example**

```
<style>
h1 {
  text-decoration-line: underline;
  text-decoration-thickness: auto; /* this is default */
}
h2 {
  text-decoration-line: underline;
  text-decoration-thickness: 5px;
}
h3 {
  text-decoration-line: underline;
  text-decoration-thickness: 25%;
}
p {
  text-decoration-line: underline;
  text-decoration-color: red;
  text-decoration-style: double;
  text-decoration-thickness: 5px;
}
</style>
</head>
<body>
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<p>A paragraph.</p>
```

Heading 1

Heading 2

Heading 3

A paragraph.

CSS Text

- **Text Transformation**
- The text-transform property is used to specify uppercase and lowercase letters in a text. This property has two values: *uppercase*, *lowercase*
- **The CSS Text Spacing Properties**

Property	Description
<u>letter-spacing</u>	Specifies the space between characters in a text
<u>line-height</u>	Specifies the line height
<u>text-indent</u>	Specifies the indentation of the first line in a text-block
<u>white-space</u>	Specifies how to handle white-space inside an element
<u>word-spacing</u>	Specifies the space between words in a text

CSS Text

▪ Text Shadow

- The text-shadow property [adds shadow to text](#). In its simplest use, you only specify the horizontal shadow (2px) and the vertical shadow (2px):

```
<!DOCTYPE html>
<html>
<head>
<style>
h1 {
  color: white;
  text-shadow: 2px 2px 4px #000000;
}
h2 {
  text-shadow: 2px 2px 5px red;
}
</style>
</head>
<body>

<h1>Text-shadow effect!</h1>
<h2>Text-shadow effect!</h1>

</body>
</html>
```

Text-shadow effect!

Text-shadow effect!

CSS Fonts

- Generic Font Families

Generic Font Family	Examples of Font Names
Serif	Times New Roman Georgia Garamond
Sans-serif	Arial Verdana Helvetica
Monospace	Courier New Lucida Console Monaco
Cursive	<i>Brush Script MT</i> <i>Lucida Handwriting</i>
Fantasy	Copperplate Papyrus

CSS Fonts

- **The CSS font-family Property**

- In CSS, we use the font-family property [to specify the font of a text.](#)

- **CSS Web Safe Fonts**

- Arial (sans-serif)
- Verdana (sans-serif)
- Tahoma (sans-serif)
- Trebuchet MS (sans-serif)
- Times New Roman (serif)
- Georgia (serif)
- Garamond (serif)
- Courier New (monospace)
- Brush Script MT (cursive)

CSS Fonts

- **Commonly Used Font Fallbacks**

- Below are some commonly used font fallbacks, organized by the 5 generic font families:
 - Serif
 - Sans-serif
 - Monospace
 - Cursive
 - Fantasy

CSS Fonts

- **Serif Fonts**

font-family	Example text
"Times New Roman", Times, serif	This is a Heading This is a paragraph.
Georgia, serif	This is a Heading This is a paragraph.
Garamond, serif	This is a Heading This is a paragraph.

CSS Fonts

▪ San-Serif Fonts

font-family	Example text
Arial, Helvetica, sans-serif	This is a Heading This is a paragraph.
Tahoma, Verdana, sans-serif	This is a Heading This is a paragraph.
"Trebuchet MS", Helvetica, sans-serif	This is a Heading This is a paragraph.
Geneva, Verdana, sans-serif	This is a Heading This is a paragraph.

CSS Fonts

- **CSS Font Style**

- The font-style property *is mostly used to specify italic text.*
- This property has three values: *normal, italic, oblique* - The text is "leaning" (oblique is very similar to italic, but less supported)

- **Font Size**

- The font-size property *sets the size of the text.*
 - *Set Font Size With Pixels:* Setting the text size with pixels gives you full control over the text size.
 - *Set Font Size With Em:* To allow users to resize the text (in the browser menu), many developers use em instead of pixels.
 - The size can be calculated from pixels to em using this formula: *pixels/16=em*

CSS Fonts

- Example

```
<style>
h1 {
  font-size: 3em;
}

h2 {
  font-size: 2em;
}

p {
  font-size: 14px;
}
</style>
</head>
<body>

<h1>3em Header</h1>
<h2>2em Header</h2>
<p>14px paragraph.</p>
```

3em Header

2em Header

14px paragraph.

CSS Fonts

■ Google Fonts

- If you do not want to use any of the standard fonts in HTML, you can use Google Fonts.
- Just add a special style sheet link in the <head> section and then refer to the font in the CSS.

```
<!DOCTYPE html>
<html>
<head>
<link rel="stylesheet" href="https://fonts.googleapis.com/css?family=Sofia">
<style>
body {
  font-family: "Sofia", sans-serif;
}
</style>
</head>
<body>

<h1>Sofia Font</h1>
<p>Lorem ipsum dolor sit amet.</p>
<p>123456790</p>

</body>
</html>
```

Sofia Font

Lorem ipsum dolor sit amet.

123456790

CSS Fonts

- **The CSS Font Property**
- To shorten the code, it is also possible to specify all the individual font properties in one property.
- The font property is a shorthand property for:
 - font-style
 - font-variant
 - font-weight
 - font-size/line-height
 - font-family

CSS Links

- Links can be styled with any CSS property (e.g. color, font-family, background, etc.).
- The four links states are:
 - **a:link** - a normal, unvisited link
 - **a:visited** - a link the user has visited
 - **a:hover** - a link when the user mouses over it
 - **a:active** - a link the moment it is clicked
- **Text Decoration:** The text-decoration property is mostly used to remove underlines from links. It is set to none.
- **Background Color:** The background-color property can be used to specify a background color for links.

CSS Links

- Example

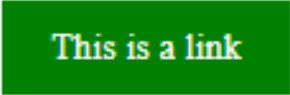
```
<!DOCTYPE html>
<html>
<head>
<style>
a:link, a:visited {
  background-color: white;
  color: black;
  border: 2px solid green;
  padding: 10px 20px;
  text-align: center;
  text-decoration: none;
  display: inline-block;
}

a:hover, a:active {
  background-color: green;
  color: white;
}
</style>
</head>
<body>
```

```
<h2>Hovered Link Button</h2>
```

```
<a href="default.asp" target="_blank">This is a link</a>
```

Hovered Link Button



This is a link

CSS Cursor Type

▪ Example

```
<!DOCTYPE html>
<html>
<body>

<h2>The cursor Property</h2>

<p>Mouse over the words to change the cursor.</p>
<span style="cursor:auto">auto</span><br>
<span style="cursor:crosshair">crosshair</span><br>
<span style="cursor:default">default</span><br>
<span style="cursor:e-resize">e-resize</span><br>
<span style="cursor:help">help</span><br>
<span style="cursor:move">move</span><br>
<span style="cursor:n-resize">n-resize</span><br>
<span style="cursor:ne-resize">ne-resize</span><br>
<span style="cursor:nw-resize">nw-resize</span><br>
<span style="cursor:pointer">pointer</span><br>
<span style="cursor:progress">progress</span><br>
<span style="cursor:s-resize">s-resize</span><br>
<span style="cursor:se-resize">se-resize</span><br>
<span style="cursor:sw-resize">sw-resize</span><br>
<span style="cursor:text">text</span><br>
<span style="cursor:w-resize">w-resize</span><br>
<span style="cursor:wait">wait</span><br>
```

The cursor Property

Mouse over the words to change the cursor.

auto
crosshair
default
e-resize
help
move
n-resize
ne-resize
nw-resize
pointer
progress
s-resize
se-resize
sw-resize
text
w-resize
wait

CSS Lists

- The CSS list properties allow you to:
 - Set [different list item markers for ordered lists](#)
 - Set [different list item markers for unordered lists](#)
 - Set an [image as the list item](#) marker
 - Add [background colors](#) to lists and list items

CSS Lists

▪ Example

```
<style>
ul {
  background: lightgray;
  padding: 20px;
  list-style-image: url('sqpurple.gif');
}
ul li {
  background: #ffffff;
  color: darkblue;
  margin: 5px;
}
ol {
  list-style-type: circle;
}
</style>
</head>
<body>
<h2>Styling Lists With Colors and Types</h2>
<ul>
  <li>Coffee</li>
  <li>Tea</li>
  <li>Coca Cola</li>
</ul>
<ol>
  <li>CSS</li>
  <li>Java</li>
  <li>C++ Cola</li>
</ol>
```

Styling Lists With Colors and Types



- Coffee
- Tea
- Coca Cola

- CSS
- Java
- C++ Cola

CSS Tables

- **Table Borders:** To specify table borders in CSS, use the border property.
- **Table Width and Height:** The width and height of a table are defined by the width and height properties.
- **Table Alignment**
 - **Horizontal Alignment:** The text-align property sets the horizontal alignment (like left, right, or center) of the content in <th> or <td>.
 - **Vertical Alignment:** The vertical-align property sets the vertical alignment (like top, bottom, or middle) of the content in <th> or <td>.
- **Collapse Table Borders:** The border-collapse property sets whether the table borders should be collapsed into a single border.s

CSS Tables

- **Table Padding:** To control the space between the border and the content in a table, use the padding property on `<td>` and `<th>` elements.
- **Horizontal Dividers:** Add the `border-bottom` property to `<th>` and `<td>` for horizontal dividers.
- **Hoverable Table:** Use the `:hover` selector on `<tr>` to highlight table rows on mouse over.
- **Table Color:** It is used to specify the background color and text color of `<th>` elements.

CSS Layout - The display Property

- **The display Property**
- The display property specifies if/how an element is displayed.
- Every HTML element has a default display value depending on what type of element it is. The default display value for most elements is block or inline.
- **Block-level Elements:** A block-level element always starts on a new line and takes up the full width available (stretches out to the left and right as far as it can).
- **Inline Elements:** An inline element does not start on a new line and only takes up as much width as necessary.
- **display: none** → is commonly used with JavaScript to hide and show elements without deleting and recreating them.

CSS Layout - The display Property

▪ Examples of block-level elements

`<div>`

`<h1> - <h6>`

`<p>`

`<form>`

`<header>`

`<footer>`

`<section>`

Examples of inline elements

`<span>`

`<a>`

`<img>`